Project Intent – QuakeShift

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Our idea

We want to build an earthquake detection system that uses open seismometer data from stations in Europe. The detection is done with existing pretrained models, so we don't train anything ourselves. The main point of the project is that the system reacts to the carbon intensity of the power grid. If the grid is "dirty", it uses a cheaper model and fewer stations. If the grid is clean, it uses the bigger model and also goes back over the data from the cheap period to find small earthquakes it might have missed.

What we plan to do

- Data: we get waveforms from GEOFON (SeedLink). For testing and the demo we want to replay a real earthquake, e.g. the M4.6 in Campi Flegrei (near Naples) on 30 June 2025.
- Models: STA/LTA as a simple trigger, and PhaseNet and EQTransformer from SeisBench.
- Architecture: one service reads the data and puts it on a message broker. Picker services run the models and find the P and S waves. An associator then checks if several stations picked something at the same time, which means it's an earthquake. Results go to an API.
- To check if our detections are correct, we compare them with the EMSC earthquake catalogue.
- Carbon-aware part: a controller reads a simulated carbon signal and switches between three modes:
  - performance: EQTransformer on all stations
  - balanced: PhaseNet on all stations
  - eco: STA/LTA first and PhaseNet only where it triggers, fewer stations
- SLOs (first draft): earthquakes from M3 detected within 60 seconds in every mode, and the queue should not keep growing.
- We measure energy and CO_2 with CodeCarbon (per station-hour and per detected earthquake).
- We run everything in Docker on a local Kubernetes cluster (kind or minikube), with GitHub Actions for CI/CD and Prometheus + Grafana for monitoring.

Questions

1. Do you think the scope is okay for two people?
2. Can we use replayed historical data for testing and the demo, next to the live data?
3. Is a local Kubernetes cluster enough, or should we deploy to a public cloud?
4. Is a simulated carbon signal enough, or should we also use real data (e.g. the Austrian grid mix from SMARD)?
5. CodeCarbon can only estimate the energy inside containers. Is that fine?
6. Are we allowed to tune things like the STA/LTA threshold, or does that count as training?